

NORTH CENTRAL STATE, RI, USA

WMO: 725054

Lat: 41.921N

Lon: 71.491W

Elev: 441

StdP: 14.46

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 64710

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.8	9.6	-11.1	3.1	8.0	-6.7	3.9	13.4	24.3	23.4	21.7	24.9	8.3	300	0.482

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	88.1	73.7	84.3	71.7	82.0	69.9	75.8	84.3	74.1	81.5	72.4	79.3	6.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.0	124.4	80.2	72.0	120.3	78.9	70.2	113.1	77.1	39.5	84.6	37.8	81.1	36.4	79.1	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.8	17.5	15.5	DB	0.7	92.6	4.1	3.2	-2.3	94.9	-4.7	96.8	-7.0	98.5	-10.0	100.8	(4)
			WB	-1.7	78.6	4.0	1.5	-4.6	79.7	-6.9	80.6	-9.1	81.4	-12.0	82.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.6	28.0	29.7	36.7	47.7	56.6	65.5	71.3	69.8	62.6	51.9	41.8	32.6
	DBStd	17.09	10.64	9.78	9.40	7.87	7.38	6.63	5.16	5.46	6.63	7.91	8.65	8.88
	HDD50	2714	685	569	426	132	16	0	0	0	1	71	273	541
	HDD65	6149	1147	987	877	521	281	76	8	17	123	412	697	1003
	CDD50	2580	3	2	14	64	221	464	660	615	378	130	26	3
	CDD65	541	0	0	1	3	21	90	203	167	50	6	0	0
	CDH74	4118	0	0	8	49	246	721	1568	1186	311	29	0	0
	CDH80	1085	0	0	2	14	73	185	442	304	63	2	0	0
Wind	WSAvg	6.6	7.8	7.8	8.1	7.7	6.8	5.4	4.9	4.8	5.3	6.1	7.1	7.4
Precipitation	PrecAvg	48.8	4.1	3.5	5.0	4.4	3.7	3.7	3.5	3.8	4.1	4.4	4.0	4.6
	PrecMax	57.7	7.1	7.8	14.8	8.1	7.8	11.5	9.1	6.5	9.3	14.1	7.0	8.1
	PrecMin	38.5	1.6	1.9	0.2	1.3	1.2	0.2	1.3	0.9	1.4	0.5	0.6	1.4
	PrecStd	5.9	1.4	1.5	3.0	1.8	1.6	3.0	1.7	1.7	2.0	3.1	1.5	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.5	60.5	72.3	81.8	87.8	90.1	91.1	90.4	87.4	78.5	67.5	61.4
		MCWB	54.8	52.5	55.9	62.9	66.4	74.7	76.1	76.8	73.8	64.3	59.4	54.9
	2%	DB	54.4	54.0	62.5	72.4	81.2	84.4	88.1	86.3	81.2	72.5	63.4	55.2
		MCWB	50.0	48.3	50.6	55.6	64.1	71.0	74.4	73.9	68.9	63.5	57.3	52.1
	5%	DB	47.9	48.0	55.1	66.0	74.8	81.6	84.3	82.6	77.4	68.9	59.7	51.9
		MCWB	43.9	42.7	46.9	53.3	61.2	69.1	72.3	71.3	67.1	61.9	53.7	48.1
	10%	DB	42.8	44.5	50.3	61.5	70.3	78.8	81.9	81.0	73.4	64.4	55.4	46.0
		MCWB	38.0	38.6	43.5	50.0	58.3	66.7	70.6	69.8	65.4	57.6	49.9	41.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.2	54.9	58.3	64.0	70.1	76.4	78.2	78.2	74.4	69.9	62.7	57.8
		MCDB	59.4	57.4	69.0	73.0	81.2	86.8	88.8	87.9	81.9	73.8	64.4	60.6
	2%	WB	50.7	48.4	53.0	59.6	66.9	73.6	76.0	75.5	71.9	66.1	58.8	53.2
		MCDB	53.7	52.7	58.8	68.9	77.6	82.6	84.8	83.2	77.9	70.0	62.4	55.3
	5%	WB	44.1	43.5	48.2	55.7	63.5	71.1	74.2	73.9	69.6	62.9	55.2	47.6
		MCDB	47.7	47.8	54.3	63.2	71.2	78.9	82.0	79.9	74.6	66.8	58.5	50.7
	10%	WB	38.5	38.6	44.0	52.4	60.0	68.8	72.5	72.2	67.5	59.3	51.3	42.1
		MCDB	42.1	43.0	50.0	59.8	68.0	75.4	79.6	78.1	72.9	64.6	55.5	46.1
Mean Daily Temperature Range	5% DB	MDBR	14.9	16.3	17.0	19.2	19.4	18.8	18.6	18.3	18.7	18.5	16.0	14.6
		MCDBR	20.4	20.9	23.5	27.1	26.4	22.9	20.9	20.0	21.2	21.3	19.7	20.1
		MCWBR	18.0	16.5	15.5	15.2	13.7	12.7	10.3	10.2	12.2	13.9	15.6	18.7
	5% WB	MCDBR	20.4	20.2	22.1	22.5	23.6	20.2	17.8	17.3	17.7	18.4	17.7	19.5
		MCWBR	19.0	17.9	17.0	15.2	13.7	12.4	10.0	9.7	11.7	14.6	16.3	19.5
Clear-Sky Solar Irradiance	taub	0.290	0.299	0.324	0.365	0.405	0.448	0.469	0.431	0.366	0.330	0.305	0.291	
	taud	2.494	2.482	2.473	2.387	2.307	2.220	2.169	2.262	2.421	2.538	2.569	2.563	
	Ebn at Noon	275	290	294	287	277	264	257	263	275	275	268	266	
	Edh at Noon	23	27	31	36	41	44	46	41	32	26	22	20	
All-Sky Solar Radiation	RadAvg	590	866	1183	1517	1672	1825	1897	1704	1383	924	636	489	
	RadStd	41	78	83	162	177	182	105	87	101	84	48	49	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (71 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+169	+81

Nomenclature: See separate page